

31/10/2023

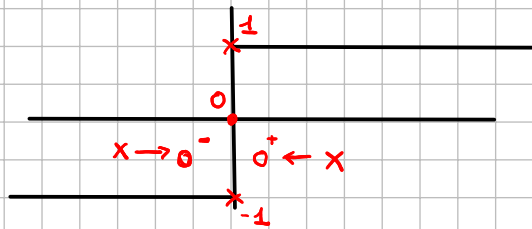
DEF (LIMITE DESTRO E SINISTRO): $A \subseteq \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACC. PER A , $f: A \rightarrow \mathbb{R}$, $l \in \mathbb{R}$, SI DICE:

$$\lim_{\substack{x \rightarrow x_0^+ \\ (x \rightarrow x_0^-)}} f(x) = l$$

SE $\forall V$ INTORNO DI l , $\exists \delta > 0$ t.c. $f(x) \in V \quad \forall x$ t.c. $\begin{matrix} x \in (x_0, x_0 + \delta) \\ 0 < x - x_0 < \delta \\ (-\delta < x - x_0 < 0) \\ x \in (x_0 - \delta, x_0) \end{matrix}$

ES:

- $\lim_{x \rightarrow 0^+} \operatorname{sgn}(x) = 1$
- $\lim_{x \rightarrow 0^-} \operatorname{sgn}(x) = -1$



$\forall \varepsilon > 0, \exists \delta > 0$ t.c.
 $|\operatorname{sgn}(x) - 1| < \varepsilon$
 $\forall x$ t.c. $0 < x < \delta$

ES: $\lim_{x \rightarrow 0^-} e^{\frac{1}{x}} = 0 \quad \forall \varepsilon > 0, \exists \delta > 0$ t.c. $e^{\frac{1}{x}} < \varepsilon$ ME $\frac{1}{x} < \log(\varepsilon)$

ME $x < 0$ $\frac{1}{\log \varepsilon} < x < 0$ $S = -\frac{1}{\log(\varepsilon)}$

$\lim_{x \rightarrow 0^+} e^{\frac{1}{x}} = +\infty \quad \forall M > 0, \exists \delta > 0$ t.c. $e^{\frac{1}{x}} > M$ ME $\frac{1}{x} > \log(M)$ ME $x > 0$ $M > 1$
 $0 < x < \frac{1}{\log(M)}$ $S = \frac{1}{\log(M)}$

PROP: $A \subseteq \mathbb{R}$, $f: A \rightarrow \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACC. PER A , $l \in \mathbb{R}$, ALLORA:

$$\lim_{x \rightarrow x_0} f(x) = l \iff \begin{cases} \lim_{x \rightarrow x_0^+} f(x) = l \\ \lim_{x \rightarrow x_0^-} f(x) = l \end{cases}$$

ES: $\exists \lim_{x \rightarrow 0} \operatorname{sgn}(x)$ DATO CHE: $\lim_{x \rightarrow 0^+} \operatorname{sgn}(x) = 1$ E $\lim_{x \rightarrow 0^-} \operatorname{sgn}(x) = -1$

TEO (UNICITÀ DEL LIMITE): $A \subseteq \mathbb{R}$, $f: A \rightarrow \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACC. PER A . ALLORA SE ESISTE IL

$\lim_{x \rightarrow x_0} f(x)$ TALE LIMITE È UNICO.

NO LIM!

TEO (CONFRONTO): $f, g: A \rightarrow \mathbb{R}$, $x_0 \in \bar{A}$ p. acc. per A , $l_1, l_2 \in \bar{\mathbb{R}}$.

SUPPONIAMO CHE:

1) $f(x) \leq g(x)$ DEFINIT. PER $x \rightarrow x_0$ 2) $\exists \lim_{x \rightarrow x_0} f(x) = l_1$, $\exists \lim_{x \rightarrow x_0} g(x) = l_2$

$$l_1 \leq l_2$$

NO NM!

N.B. $f(x) < g(x)$ NON È VERO CHE $l_1 < l_2$. ES: $f(x) = x^2$ $g(x) = 0$

$$g(x) < f(x) \text{ DEF. PER } x \rightarrow 0$$

$$\text{MA } \lim_{x \rightarrow 0} f(x) = 0 = \lim_{x \rightarrow 0} g(x)$$

TEO (DEI 2 CARABINIERI): $A \subseteq \mathbb{R}$, x_0 PUNTO DI ACC. DI A , $f, g, h: A \rightarrow \mathbb{R}$ t.c.

1) $f(x) \leq g(x) \leq h(x)$ DEF. PER $x \rightarrow x_0$ 2) $\exists \lim_{x \rightarrow x_0} f(x) = l = \lim_{x \rightarrow x_0} h(x)$ $l \in \mathbb{R}$

ALLORA: $\exists \lim_{x \rightarrow x_0} g(x) = l$

NM: $\forall \varepsilon > 0 \exists V_1, V_2$ INTORNI DI x_0 t.c.:

$$x \in A \cap V_1 \setminus \{x_0\} \Rightarrow |f(x) - l| < \varepsilon \Leftrightarrow l - \varepsilon < f(x) < l + \varepsilon$$

$$x \in A \cap V_2 \setminus \{x_0\} \Rightarrow |h(x) - l| < \varepsilon \Leftrightarrow l - \varepsilon < h(x) < l + \varepsilon$$

$\exists V_3$ INTORNO DI x_0 t.c.:

$$x \in A \cap V_3 \setminus \{x_0\} \Rightarrow f(x) \leq g(x) \leq h(x)$$

PRENDIAMO $U = V_1 \cap V_2 \cap V_3 \Rightarrow$ INTORNO SEMPRE DI x_0

$$\text{SE } x \in U \cap A \setminus \{x_0\} \Rightarrow l - \varepsilon < f(x) \leq g(x) \leq h(x) < l + \varepsilon$$

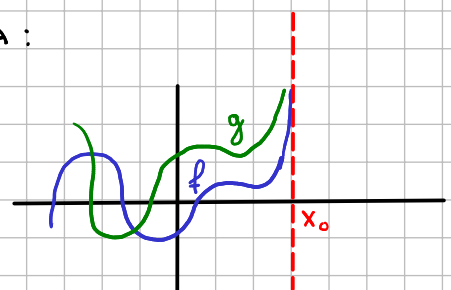
$$\text{CI O È: } l - \varepsilon < g(x) < l + \varepsilon \Rightarrow |g(x) - l| < \varepsilon$$

TEO (SINGOLO CARABINIERE): SIANO $f, g: A \rightarrow \mathbb{R}$, $A \subseteq \mathbb{R}$, $x_0 \in \mathbb{R}$ PUNTO DI ACC. DI A .

SUPPONIAMO CHE $f(x) \leq g(x)$ DEF. PER $x \rightarrow x_0$ ALLORA:

1) SE $\lim_{x \rightarrow x_0} f(x) = +\infty \Rightarrow \lim_{x \rightarrow x_0} g(x) = +\infty$

2) SE $\lim_{x \rightarrow x_0} g(x) = -\infty \Rightarrow \lim_{x \rightarrow x_0} f(x) = -\infty$



NO DIM!

ES: $\lim_{x \rightarrow +\infty} \frac{[x]}{x}$ $\rightarrow$ PARTE INTERA $\Rightarrow [2,73] \rightarrow 2$

QUINDI DI SICURO: $[x] \leq x$

$[1] = 1$
PER QUESTO $\leq$

TEO DEI 2 CARABINIERI:

E DI SICURO: $[x] > x - 1$

$\hookrightarrow$ LA POSSIAMO SCRIVERE COME: $[x] \geq x - 1$

$\frac{x-1}{x} \leq \frac{[x]}{x} \leq \frac{x}{x}$
 $\underbrace{\quad}_f \quad \underbrace{\quad}_g \quad \underbrace{\quad}_h$

$\hookrightarrow \lim_{x \rightarrow +\infty} \frac{x-1}{x} = \lim_{x \rightarrow +\infty} 1 - \frac{1}{x} = 1 \Rightarrow \lim_{x \rightarrow +\infty} \frac{[x]}{x} = 1$

TEO (LIMITE DELLA SOMMA - CASO FINITO): $f, g: A \rightarrow \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACCUH. PER A, SUPPONIAMO:

$\lim_{x \rightarrow x_0} f(x) = l_1 \in \mathbb{R}$ e $\lim_{x \rightarrow x_0} g(x) = l_2 \in \mathbb{R}$ ALLORA:

$\lim_{x \rightarrow x_0} (f(x) + g(x)) = \lim_{x \rightarrow x_0} f(x) + \lim_{x \rightarrow x_0} g(x) = l_1 + l_2$

DM: $\forall \varepsilon > 0, \exists V_1, V_2$ INTORNI DI x_0 L.C.

$x \in A \cap V_1 \setminus \{x_0\} \Rightarrow |f(x) - l_1| < \frac{\varepsilon}{2} \xrightarrow{\text{POSSO SCEGLIERE QUALUNQUE } \varepsilon} \Leftrightarrow \frac{\varepsilon}{2} - l_1 < f(x) < \frac{\varepsilon}{2} + l_1$

$x \in A \cap V_2 \setminus \{x_0\} \Rightarrow |g(x) - l_2| < \frac{\varepsilon}{2} \Leftrightarrow \frac{\varepsilon}{2} - l_2 < g(x) < \frac{\varepsilon}{2} + l_2$

SE $x \in V_1 \cap V_2 \cap A \setminus \{x_0\}$ ALLORA:

$l_1 - \frac{\varepsilon}{2} + l_2 - \frac{\varepsilon}{2} < f(x) + g(x) < l_1 + \frac{\varepsilon}{2} + l_2 + \frac{\varepsilon}{2}$

$l_1 + l_2 - \varepsilon < f(x) + g(x) < l_1 + l_2 + \varepsilon$

$\text{cio } \varepsilon \Rightarrow |f(x) + g(x) - (l_1 + l_2)| < \varepsilon \Rightarrow \lim_{x \rightarrow x_0} f(x) + g(x) = l_1 + l_2$

TEO (LIMITE DELLA SOMMA - CASO INFINITO): $f, g: A \rightarrow \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACC. PER A:

1) $\lim_{x \rightarrow x_0} f(x) = +\infty$ ($0 - \infty$) 2) $g(x)$ È INFERIORM. LIMITATA (SUPERIOR. LIMITATA)

ALLORA IL $\lim_{x \rightarrow x_0} f(x) + g(x) = +\infty$ ($-\infty$)

$\lim_{x \rightarrow +\infty} - \frac{g_{oon}^{104} + ANNAL^{2/3} - SAJIB}{\frac{1}{4} g_{oon}^{104} - MOSTRO} = -4$

ES: $f(x) = x, g(x) = -x$ $\lim_{x \rightarrow +\infty} x = +\infty, \lim_{x \rightarrow -\infty} -x = -\infty \Rightarrow \lim_{x \rightarrow +\infty} f(x) + g(x) =$

$= \lim_{x \rightarrow +\infty} \overset{0}{x-x} = 0$

$f(x) = x, g(x) = -x + \overset{-1 \circ 1}{\cos(x)}$ $\lim_{x \rightarrow +\infty} x = +\infty, \lim_{x \rightarrow -\infty} -x + \cos(x) = -\infty$ $\nexists \lim_{x \rightarrow +\infty} f(x) + g(x) = \lim_{x \rightarrow +\infty} \cos(x)$

TEO (LIMITE DEL PRODOTTO): $f, g: A \rightarrow \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACC. PER A , SUPPONIAMO CHE:

$$\lim_{x \rightarrow x_0} f(x) = l_1 \in \mathbb{R} \quad \text{e} \quad \lim_{x \rightarrow x_0} g(x) = l_2 \in \mathbb{R} \quad \text{ALLORA:}$$

$$\lim_{x \rightarrow x_0} (f(x) \cdot g(x)) = \lim_{x \rightarrow x_0} f(x) \cdot \lim_{x \rightarrow x_0} g(x) = l_1 \cdot l_2$$

DIM NO !!

TEO (LIMITE DEL QUOZIENTE): $f, g: A \rightarrow \mathbb{R}$, $x_0 \in \overline{\mathbb{R}}$ PUNTO DI ACC. PER A , SUPPONIAMO CHE:

$$\lim_{x \rightarrow x_0} f(x) = l_1 \in \mathbb{R} \quad \text{e} \quad \lim_{x \rightarrow x_0} g(x) = l_2 \in \mathbb{R} \quad \text{ALLORA:}$$

$$\lim_{x \rightarrow x_0} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow x_0} f(x)}{\lim_{x \rightarrow x_0} g(x)} = \frac{l_1}{l_2}$$

QUALCHE ALTRA REGOLA:

1) " $\frac{1}{\infty} = 0$ " SE $f \in \text{LIMITATA}$ E $|g(x)| = +\infty$ ALLORA $\frac{f(x)}{g(x)} \xrightarrow{x \rightarrow x_0} 0$

2) " $\frac{1}{0} = \infty$ " SE $f > 0$ CIOE' $\exists c$ t.c. $|f(x)| \geq c$ $c \neq 0$ E $g(x) \xrightarrow{x \rightarrow x_0} 0$ ALLORA

$$\left| \frac{f(x)}{g(x)} \right| \xrightarrow{x \rightarrow x_0} +\infty$$

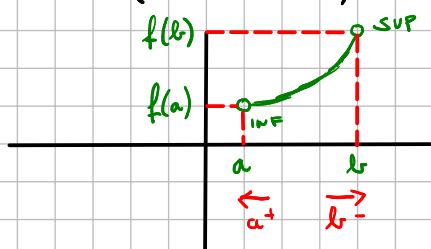
3) " $1 \cdot \infty = \infty$ " SE $f \Rightarrow |f(x)| \geq c$ $c \neq 0$ E $g(x) \xrightarrow{x \rightarrow x_0} +\infty$ ALLORA:

$$|f(x) \cdot g(x)| \xrightarrow{x \rightarrow x_0} +\infty$$

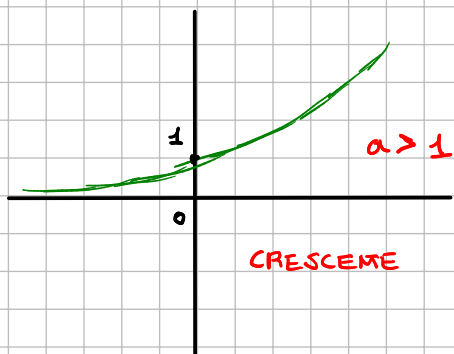
TEO (LIMITE DELLE FUNZIONI MONOTONE): SIANO $a, b \in \overline{\mathbb{R}}$, $a < b$. $f: (a, b) \rightarrow \mathbb{R}$ SI HA CHE:

• SE f E' CRESCENTE SU (a, b) , ALLORA: $\exists \lim_{x \rightarrow b^-} f(x) = \sup f := \sup \{ f(x) : x \in (a, b) \}$
(DECRESCENTE)

$$\exists \lim_{x \rightarrow a^+} f(x) = \inf f := \inf \{ f(x) : x \in (a, b) \}$$

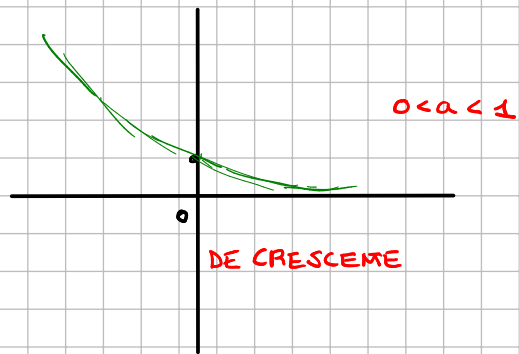


ES: ESPONENZIALE: SIA $a > 0$, $a \neq 1$



$$\lim_{x \rightarrow +\infty} a^x = \sup_{x \in \mathbb{R}} a^x = +\infty$$

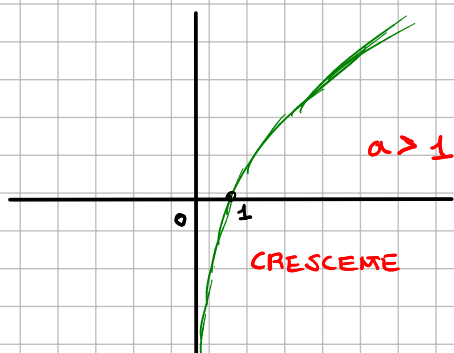
$$\lim_{x \rightarrow -\infty} a^x = \inf_{x \in \mathbb{R}} a^x = 0$$



$$\lim_{x \rightarrow +\infty} a^x = \sup_{x \in \mathbb{R}} a^x = 0$$

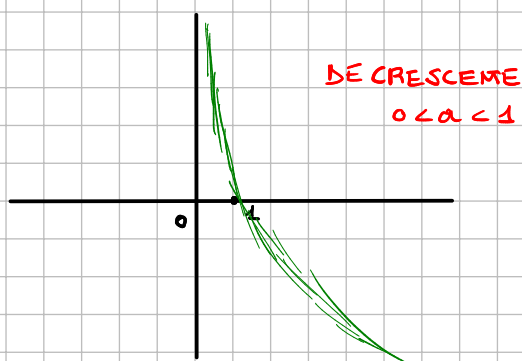
$$\lim_{x \rightarrow -\infty} a^x = \inf_{x \in \mathbb{R}} a^x = +\infty$$

• LOGARITMO: $a > 0, a \neq 1$:



$$\lim_{x \rightarrow +\infty} a^x = \sup_{x \in \mathbb{R}} \log_a x = +\infty$$

$$\lim_{x \rightarrow 0^+} a^x = \inf_{x \in \mathbb{R}} \log_a x = -\infty$$



$$\lim_{x \rightarrow +\infty} a^x = \sup_{x \in \mathbb{R}} \log_a x = -\infty$$

$$\lim_{x \rightarrow 0^+} a^x = \inf_{x \in \mathbb{R}} \log_a x = +\infty$$